

Mathematical Principles of Large Language Models

Transformers, Attention, GPT, BERT, and a Python Demonstration

Diogo Santos Matěj Polák

Mathematics for Machine Learning
Instituto Superior Técnico

Core Idea of Self-Attention

Self-attention asks:

When processing one token, how much should it attend to every other token?

Each token embedding X is projected into three vectors:

$$Q = XW_Q,$$

$$K = XW_K,$$

$$V = XW_V.$$

- **Query:** what this token is looking for.
- **Key:** what this token offers as a match.
- **Value:** information passed forward if attended to.

Scaled Dot-Product Attention

For tokens i and j , the raw attention score is

$$a_{ij} = q_i^\top k_j.$$

(Compare to cosine similarity!)

Computing all pairs gives

$$A = QK^\top.$$

The final attention output is

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V.$$

Scores are scaled to avoid large dot products when d_k is large.

Softmax converts scores into a probability distribution.

Causal Masking

In autoregressive models, future tokens must not influence earlier tokens.

Illegal future connections are set to $-\infty$ before softmax:

$$\text{Mask}(\tilde{S})_{ij} = \begin{cases} \tilde{S}_{ij}, & j \leq i, \\ -\infty, & j > i. \end{cases}$$

Since

$$e^{-\infty} = 0,$$

those entries become zero after softmax.

Multi-Head Attention

A single attention head captures only one type of token relationship.

Multi-head attention runs several attention heads in parallel:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) W^O.$$

Each head is

$$\text{head}_i = \text{Attention}(QW_Q, KW_i^K, VW_i^V).$$

Intuition: different heads can specialize in different relations, such as syntax, semantics, or coreference.

Positional Encoding

Attention alone is permutation-invariant: it does not know token order.

Transformers add positional encodings to token embeddings:

$$PE_{(pos, 2i)} = \sin\left(\frac{pos}{10000^{2i/d_{model}}}\right),$$
$$PE_{(pos, 2i+1)} = \cos\left(\frac{pos}{10000^{2i/d_{model}}}\right).$$

This gives the model information about absolute and relative position.

GPT

- Pre-training + fine-tuning paradigm.
- Transformer decoder.
- Autoregressive left-to-right modeling.
- Objective: next-token prediction.
- Naturally suited for generation.

$$P(u_i \mid u_{i-k}, \dots, u_{i-1})$$

BERT

- Pre-training + fine-tuning paradigm.
- Transformer encoder.
- Bidirectional contextual representations.
- Objective: masked language modeling.
- Naturally suited for understanding tasks.

$$P(x_i \mid X_{\setminus M})$$

App: `https://2026-m4ml-2.streamlit.app`

References

-  Vaswani et al. (2017). *Attention Is All You Need*.
-  Radford and Narasimhan (2018). *Improving Language Understanding by Generative Pre-Training*.
-  Devlin et al. (2019). *BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding*.
-  Sanderson (2024). *Visualizing Attention, a Transformer's Heart*. 3Blue1Brown.
-  Starmer, J. (2023) Attention for Neural Networks, Clearly Explained!!! *StatQuest with Josh Starmer*. Available at: <https://www.youtube.com/watch?v=PSs6nxngL6k> (Accessed: May 10, 2026).

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Thank you!